

## Task 4. The Three Little Pigs

### Available marks: 23

It's a little-known fact that before the Three Little Pigs' celebrated and unfortunate encounter with the Big Bad Wolf, they encoded their housing construction plans to prevent the wolf from discovering the flaws in their design. They first divided the plan image exactly down the middle, producing a sequence of colourants  $A$  from the left half and  $B$  from the right half. A colourant is the red, green or blue element of a pixel, and in the original image each is represented by an 8-bit value (decimal 0 to 255).

- The first little pig encoded the images by calculating for each colourant in position  $i$  the value  $S_i = \text{straw}(A_i, B_i)$ . The straw function returns the sum of its arguments.
- The second little pig encoded the images by calculating for each colourant in position  $i$  the value  $D_i = \text{sticks}(A_i, B_i)$ . The sticks function returns the difference between its arguments ( $A_i - B_i$ ) plus 256 to avoid negative values.
- The third little pig applied a bricks function to the 9-bit sum and 9-bit difference values to generate three new 6-bit colourants, producing a new scrambled image.

Using ordinary arithmetic, the new colourants are calculated thus:

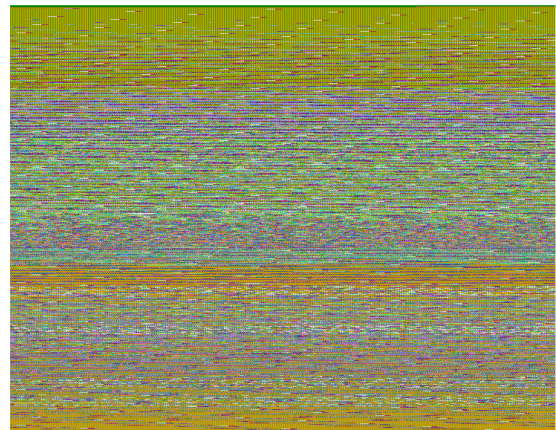
$$\text{bricks}(S_i, D_i, \text{red}) = S_i \bmod 64$$

$$\text{bricks}(S_i, D_i, \text{green}) = (S_i / 64) + 8 * (D_i / 64)$$

$$\text{bricks}(S_i, D_i, \text{blue}) = D_i \bmod 64$$

where  $/$  is the integer division operator that discards any remainder and **mod** is the remainder operator.

Thus each pair of pixels in the original image produces 3 pixels in the encoded image.



### Your Task

The resulting image is shown above and is available in the file **piggy.ppm**. It's 735 by 576 pixels. Note that it uses 6-bit data values, where the original image uses 8 bits. The original image has landscape format and is exactly 1.6 times as wide as it is high. Do the Big Bad Wolf a favour and decode it for him.

Smaller files **symbol1.ppm** and **symbol2.ppm** producing images with the same 1.6 aspect ratio are also included in the test data sets. Your program must work out the dimensions of the final image from inspecting the file header itself. You can use either arithmetic or bitwise operations to reassemble the data.

### Assessment

- Decoding and reassembling images accurately is worth **17 marks**. You must reproduce the main image but may be asked to show other test images as well.
- In addition, **3 marks** are reserved for producing a completely valid P3 file format, including header values, no colourants outside the legal range, and lines no longer than 70 characters.
- A further **3 marks** are available if you can decode by any means the TLPs' copyright notice, which is obtained by interpreting the last 83 colourants (if they have been restored exactly) as ASCII character codes.